

Experiment - 7

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Q1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

AB→C, C→D, D→A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:

Given,

$AB \rightarrow C, C \rightarrow D, D \rightarrow A$

B never appears on RHS so B must be a part of candidate keys

Finding closures,

$B^+ = B$

$AB^+ = ABCD$

$BC^+ = BCDA$

$BD^+ = BDAC$

Candidate Keys: {AB, BC, BD}

Prime Attributes: {A, B, C, D}

Non-Prime Attributes: {}

Checking FDs for BCNF,

$AB \rightarrow C$ (AB is a super key)

$C \rightarrow D$ (C is not a super key so it violates BCNF property)

And since all attributes are prime attributes, relation is in 3NF.

Q2. Relation R(ABCDE) having functional dependencies as:

A→D, B→A, BC→D, AC→BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:

Given,

$A \rightarrow D$

$B \rightarrow A$

$BC \rightarrow D$

$AC \rightarrow BE$

C never appears on RHS so C must be a part of the candidate key

Finding closures:

$C^+ = C$

$AC^+ = ACBED$

$BC^+ = BCADE$

$CD^+ = CD$

$CE^+ = CE$

Candidate Keys: {AC, BC}

Prime Attributes: {A, B, C}

Non-Prime Attributes {D, E}

Checking FDs for BCNF,

$A \rightarrow D$ (A is not a super key, hence violates property for BCNF)

Checking FDs for 3NF,

$A \rightarrow D$ (Neither A is super key nor is D a prime attribute, hence violates 3NF)

Checking FDs for 2NF,

$A \rightarrow D$ (A is a proper subset of candidate key AC and D is a non-prime attribute, hence violates 2NF)

So, it is in 1NF.

Q3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

$B \rightarrow A, A \rightarrow C, BC \rightarrow D, AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:

Given,

$B \rightarrow A$

$A \rightarrow C$

$BC \rightarrow D$

$AC \rightarrow BE$

$A^+ = ACBED$

$B^+ = BACED$

Candidate Keys: $\{A, B\}$

Prime Attributes: $\{A, B\}$

Non-Prime Attributes: $\{C, D, E\}$

Checking FDs for BCNF,

$B \rightarrow A$ (B is a super key)

$A \rightarrow C$ (A is a super key)

$BC \rightarrow D$ (BC is a super key)

$AC \rightarrow BE$ (AC is a super key)

So the table is in BCNF.

Q4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

$A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:

Given,

$A \rightarrow BCD$

$BC \rightarrow DE$

$B \rightarrow D$

$D \rightarrow A$

F never appears on RHS, so F must be a part of the candidate key

Finding closures,

$$F^+ = F$$

$$AF^+ = ABCDEF$$

$$BF^+ = BFDACE$$

$$CF^+ = CF$$

$$DF^+ = DFABCE$$

$$EF^+ = EF$$

Candidate Keys: {AF, BF, DF}

Prime Attributes: {A, B, D, F}

Non-Prime Attributes: {C, E}

Checking FDs for BCNF,

$A \rightarrow BCD$ (A is not a super key)

Checking FDs for 3NF,

$A \rightarrow BCD$ (A is not a super key and neither are C is non-prime)

Checking FDs for 2NF,

$A \rightarrow BCD$ (A is a subset of candidate key AF)

So highest NF is 1NF.

Q5. Relation R(ABCDE) having dependencies as:

$CE \rightarrow D$

$D \rightarrow B$

$C \rightarrow A$

Highest normal form? C.K set?

Solution:

Given,

$$CE \rightarrow D$$

$$D \rightarrow B$$

$$C \rightarrow A$$

C and E never appear on RHS so CE must be a part of candidate key

Finding closures,

$$CE^+ = CEDBA$$

Candidate Keys: {CE}

Prime Attributes: {C, E}

Non-Prime Attributes: {A, B, D}

Checking BCNF,

$CE \rightarrow D$ (CE is a super key)

$D \rightarrow B$ (D is not a super key)

Checking 2NF

$C \rightarrow A$ (C is a proper subset of candidate key CE, and A is non-prime, hence violates 2NF)

Highest Normal Form: 1NF

Q6. Relation R(ABCDEF) having dependencies as:

$AB \rightarrow C$

$DC \rightarrow AE$

$E \rightarrow F$

Highest normal form? C.K set?

Solution:

Given,

$AB \rightarrow C$

$DC \rightarrow AE$

$E \rightarrow F$

B and D never appear on RHS, so both B and D must be part of every candidate key.

Finding closures,

$BD^+ = BD$

$ABD^+ = ABCDEF$

$BCD^+ = ABCDEF$

Candidate Keys: {ABD, BCD}

Prime Attributes: {A, B, C, D}

Non-Prime Attributes: {E, F}

Checking BCNF,

$AB \rightarrow C$ (AB is not a super key)

Checking 3NF,

$DC \rightarrow AE$ (DC is not a super key and E is non-prime)

Checking 2NF,

$DC \rightarrow E$ (DC subset of candidate key BCD)

Highest NF: 1NF

Q7. Relation R(ABCDEFGHI) having dependencies as:

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

Highest normal form? C.K set?

Solution:

Given,

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

A, B, D never appear on RHS, so A, B and D must be part of every candidate key

$ABD^+ = ABCDEFGHI$

Candidate Keys: {ABD}

Prime Attributes: {A, B, D}

Non-Prime Attributes: {C, E, F, G, H, I}

Checking BCNF,

$AB \rightarrow C$ (AB is not a super key)

Checking 3NF,

$AB \rightarrow C$ (AB is not a super key and C is non-prime)

Checking 2NF,

$AB \rightarrow C$ (AB is a proper subset of candidate key ABD and C is non-prime)

Highest NF: 1NF

Q8. Relation R(ABCDE) having dependencies as:

AB $\rightarrow$ CD

D $\rightarrow$ A

BC $\rightarrow$ DE

Highest normal form? C.K set?

Solution:

Given,

AB $\rightarrow$ CD

D $\rightarrow$ A

BC $\rightarrow$ DE

Since B never appears on RHS, B must be part of every candidate key.

Finding closures,

B⁺ = B

AB⁺ = ABCDE

BD⁺ = ABCDE

BC⁺ = ABCDE

Candidate Keys: {AB, BD, BC}

Prime Attributes: {A, B, C, D}

Non-Prime Attributes: {E}

Checking BCNF,

D $\rightarrow$ A (D is not a super key)

Checking 3NF,

D $\rightarrow$ A (D is not a super key, but A is prime)

Highest NF: 3NF

Q9. Relation R(ABCDE) having dependencies as:

BC $\rightarrow$ ADE

D $\rightarrow$ B

Highest normal form? C.K set?

Solution:

Given,

$BC \rightarrow ADE$

$D \rightarrow B$

C, and E never appear on RHS, so C and E must be part of every candidate key.

Finding closures,

$C^+ = C$

$BC^+ = ABCDE$

$CD^+ = ABCDE$

Candidate Keys: {BC, CD}

Prime Attributes: {B, C, D}

Non-Prime Attributes: {A, E}

Checking BCNF,

$D \rightarrow B$ (D is not a super key)

Checking 3NF,

$D \rightarrow B$ (D is not a super key, but B is prime)

$BC \rightarrow ADE$ (BC is a super key)

Highest NF: 3NF

Q10. Relation R(ABC) having dependencies as:

A $\rightarrow$ B

B $\rightarrow$ C

C $\rightarrow$ A

Highest normal form? C.K set?

Solution:

Given,

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Finding closures,

$A^+ = ABC$

$B^+ = ABC$

$$C^+ = ABC$$

Candidate Keys: {A, B, C}

Prime Attributes: {A, B, C}

Non-Prime Attributes: {}

All are prime attributes so BCNF

Highest NF: BCNF

Q11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 → F3

F2 → F4

(F1, F2) → F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Solution:

Given,

$F1 \rightarrow F3$

$F2 \rightarrow F4$

$(F1, F2) \rightarrow F5$

F1 and F2 never appear on RHS, so they must be part of every candidate key.

Finding closures,

$F1F2^+ = F1F2F3F4F5$

Candidate Key: {F1F2}

Prime Attributes: {F1, F2}

Non-Prime Attributes: {F3, F4, F5}

Checking BCNF,

$F1 \rightarrow F3$ (F1 is not a super key)

Checking 3NF,

$F1 \rightarrow F3$ (F1 is not a super key and F3 is non-prime)

Checking 2NF,

$F1 \rightarrow F3$ ($F1$ is a proper subset of candidate key $F1F2$ and $F3$ is non-prime)

Highest NF: 1NF

Q12. Consider a relation $R(ABCDEFGHIJ)$ having FDs as:

$AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.

Apply the decomposition if needed to remove the redundancy.

Identify how many new tables will be formed after the decomposition.

Solution:

Given,

$AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

A, B and D never appear on RHS, so A, B and D must be part of every candidate key

Finding closures,

$ABD^+ = ABCDEFGHIJ$

Candidate Key: $\{ABD\}$

Prime Attributes: $\{A, B, D\}$

Non-Prime Attributes: $\{C, E, F, G, H, I, J\}$

Checking BCNF,

$AB \rightarrow C$ (AB is not a super key, so relation violates BCNF)

Checking 3NF,

$AB \rightarrow C$ (AB is not a super key and C is non-prime, so violates 3NF)

Checking 2NF,

$AB \rightarrow C$ (AB is a proper subset of candidate key ABD and C is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition to remove redundancy

Since the relation violates 2NF, decompose the partial dependencies:

$R_1(A, B, C)$ for $AB \rightarrow C$

$R_2(A, D, G, H)$ for $AD \rightarrow GH$

$R_3(B, D, E, F)$ for $BD \rightarrow EF$

$R_4(A, I)$ for $A \rightarrow I$

We also need a relation containing the candidate key:

$R_5(A, B, D)$ for candidate key

Since $H \rightarrow J$ is a transitive dependency, if we also want to remove that redundancy and reach 3NF, create:

$R_6(H, J)$ for $H \rightarrow J$

So, 6 tables are formed.

Q13. Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows:

$A \rightarrow B$

$B \rightarrow E$

$C \rightarrow D$

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.

Apply the decomposition if needed to remove the redundancy.

Identify how many new tables will be formed after the decomposition.

Solution:

Given,

$A \rightarrow B$

$B \rightarrow E$

$C \rightarrow D$

A and C never appear on RHS, so A and C must be part of every candidate key.

Finding closures,

$AC^+ = ABCDE$

Candidate Key: {AC}

Prime Attributes: {A, C}

Non-Prime Attributes: {B, D, E}

Checking BCNF,

$A \rightarrow B$ (A is not a super key, so it violates BCNF)

Checking 3NF,

$A \rightarrow B$ (A is not a super key and B is non-prime, so it violates 3NF)

Checking 2NF,

$A \rightarrow B$ (A is a proper subset of candidate key AC and B is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition

To remove the partial and transitive dependencies, decompose as:

R1(A, B) for $A \rightarrow B$

R2(B, E) for $B \rightarrow E$

R3(C, D) for $C \rightarrow D$

R4(A, C) for Candidate key AC

Number of new tables formed: 4 tables